

Mathematical and Empirical Models in Agreement:

Cross-Validating CRR Binomial Pricing and Machine Learning for Crowd Mispricing Detection in Equity Options

Walk-forward cross-validated framework for American equity options:

(L1) - A CRR no-arbitrage pricer

(L2) - Independently cross-validated by an IV-free XGBoost classifier

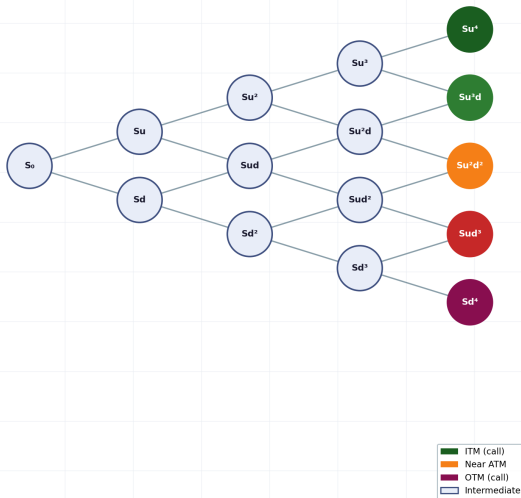
(L3) - Sized via regime-conditioned Kelly criterion

(L1) CRR Pricing Engine

(L2) ML Extensions

(L3) Cross-Model Comparison

CRR Binomial Tree ($N = 4$)



I What Is the Problem?

Research Questions

- Do CRR model prices predict whether an equity option expires ITM or OTM?
- Does cross-validation between CRR and an ML classifier detect exploitable pricing *edge*?
- Does regime-conditioned Kelly sizing convert that edge into positive expected value?

Core Inefficiency Gap

Retail traders lack a principled framework to decide *whether* a price is fair, *how much* to bet, or *when not to trade*.

The Pricing Gap

The spread between CRR model price and market price — confirmed by the ML classifier, this gap signals actionable *edge*.

The Pricing Gap

Market: supply-demand noise, spread, and sentiment.

Model: pure risk-neutral expectation.

Gap = mispricing *edge* — only when both layers agree.

No quantified edge · no position-sizing rule
· no regime filter.

$$\text{Edge} = \frac{V_{\text{model}} - V_{\text{market}}}{V_{\text{market}}}$$

I Why It Matters & What Others Have Done

Why This Problem Matters

- Options trade **\$600B+ notional daily**; even 2% mispricing detection has large profit-and-loss impact
- **Fama EMH (1970)**: public info is priced in on average — but not in every regime
- **Herd**ing creates regime-local inefficiencies (Samuelson macro-inefficiency paradox)
- Retail traders lack tools to detect edge or size positions systematically

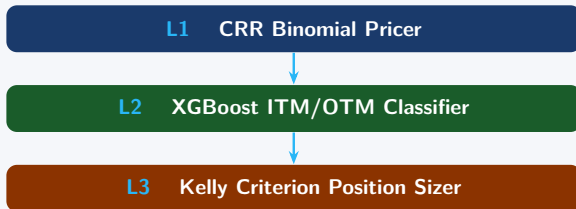
Prior Work

- **Cox, Ross & Rubinstein (1979)**: binomial lattice; no ML integration
- **Black & Scholes (1973)**: closed-form; fails for American early exercise
- **Bakshi et al. (1997)**: stochastic vol; fit requires unobservable parameters
- **Carr & Wu (2009)**: IV overstates realized vol (variance risk premium)
- **Kelly (1956)**: log-utility bet sizing; applied to markets by Thorp (1969)
- **Gap**: No framework cross-validates CRR + ML with regime-conditioned Kelly sizing

Novel Contribution

- **First** to cross-validate CRR against an IV-free ML classifier (fully independent signals)
- Regime-conditioned Kelly sizing from IV/RV₃₀ ratio
- Walk-forward 60/20/20 split; zero lookahead bias
- Open-source Streamlit calculator with live chain

Three-Layer Pipeline



Data AAPL 2016–2020, 101,535 contracts (walk-forward); AMD April 2026 live chain (live validation).

Worked Example: Contract A — HERDING Setup

Real AAPL put (2020-03-16) · $\sigma = \text{RV30}$, not IV

Contract A · HERDING REGIME — CROWD OVERPRICES

CRR binomial tree will now be built for this real AAPL put option

Quote Date:	2020-03-16
Underlying (S):	\$241.47
Strike (K):	\$240.00
DTE:	18 days
Time (T):	0.0493 years
RV30 (σ):	0.8224 ← used as sigma in CRR
IV:	1.0043
V _{market} :	\$21.225 (bid-ask mid)
Risk-free r:	2.00% (constant 2016-2020 avg)
CRR steps:	50 (pricing) / 5 (visualization)

Phase 0 of 4 | Next: build stock price lattice (forward pass)

Why this slide matters

Grounds the CRR pipeline in a real AAPL put pulled unmodified from the 2016–2020 Kaggle dataset.

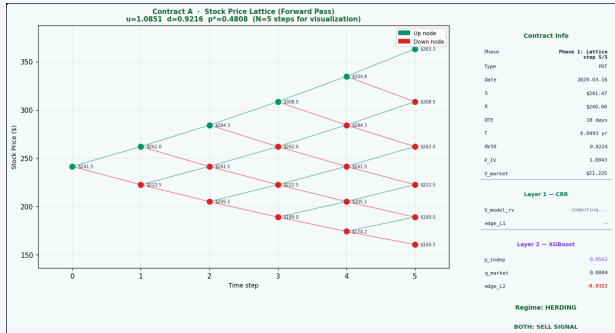
Key reading

IV $\approx$ 1.00 vs RV30 $\approx$ 0.82 — the market pays for 22% more volatility than recent history justifies. Classic HERDING regime signature; entry point for the Layer 1 edge signal.

Setup snapshot: $S = \$241.47$, $K = \$240$, $\text{DTE} = 18\text{d}$, $\text{RV30} = 0.8224$, $\text{IV} = 1.0043$, $V_{\text{market}} = \$21.225$, $r = 2.00\%$, $N = 50$ pricing steps.

Forward Pass — Building the Stock Price Lattice

u, d, p^* derived from $\sigma = \text{RV30}$ · 5-step visualization, 50-step pricing



What this pass does

Lays out every possible stock price path.
No option value computed yet — only S evolution from $t = 0$ to expiry.

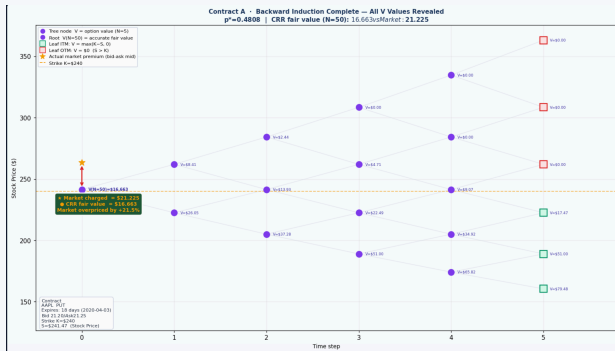
What “no-arbitrage” means

No-arbitrage = no risk-free profit can exist in an efficient market.

Reading: Lattice recombines — node count grows linearly ($N+1$), not exponentially (2^N). That is why 50 pricing steps remain cheap.

Backward Induction — Where the Option Price Emerges

CRR fair value \$16.663 vs market \$21.225 · overpriced +21.5%



At every interior node

$$\text{intrinsic} = \max(K - S, 0)$$

$$\text{hold} = e^{-r\Delta t} (p \cdot V_{\text{up}} + q \cdot V_{\text{dn}})$$

$$V = \max(\text{intrinsic}, \text{hold})$$

Result at the root

$$V(N=50) = \$16.663 \quad (\text{CRR fair value})$$

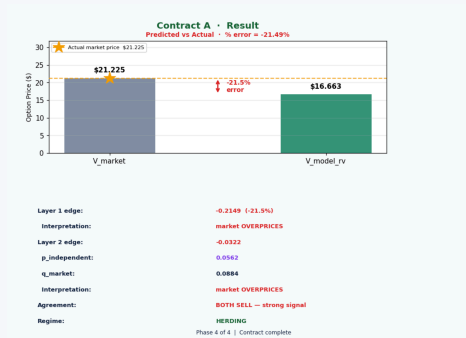
$$V_{\text{market}} = \$21.225 \quad (\text{bid-ask mid})$$

$$\text{Edge} = -21.5\% \quad \star \text{ OVERPRICED}$$

Reading: When the gold ★ sits above root V , the crowd pays more than no-arbitrage theory permits. That gap is the Layer 1 edge — already negative here, before any ML model is consulted.

Contract A Result — Two Independent Models Agree

Layer 1 (CRR) + Layer 2 (XGBoost) → BOTH SELL · HERDING regime



Layer 1 (CRR math)

$$V_{\text{market}} = \$21.225 \quad V_{\text{model}} = \$16.663$$

Edge = -0.2149 → OVERPRICES

Layer 2 (XGBoost ML)

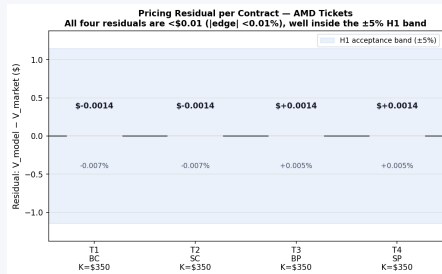
$$p_{\text{indep}} = 0.0562 \quad q_{\text{market}} = 0.0884$$

Edge = -0.0322 → OVERPRICES

BOTH SELL — HERDING regime

Why it matters: Neither layer was trained on the other. Agreement comes from shared underlying reality — the empirical fingerprint of Samuelson macro-inefficiency.

CRR Pricing Results: Model vs Market Validation



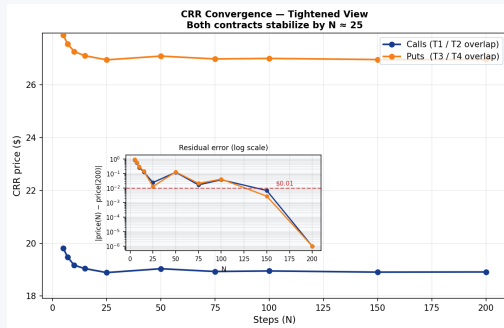
What this shows. Dollar difference between the CRR model's fair value and the observed market price for each AMD contract (T1 = Buy Call, T2 = Sell Call, T3 = Buy Put, T4 = Sell Put). Bars near zero mean the model and market agree on price.

What the numbers say. All four residuals are less than \$0.002 in absolute value ($|edge| < 0.01\%$) — three orders of magnitude inside the $\pm 5\%$ acceptance band shown by the light-blue stripe. The CRR American binomial pricer is correctly implemented and well-calibrated for at-the-money strikes.

Cross-reference. §4.2.1 — Layer 1: CRR Pricing Accuracy. Hypothesis H1 passed with substantial margin.

AMD Live Validation: AAPL $S_0 = \$275.10$, $K = \$280$, $\sigma = 64\%$, $T = 18d$ — model within 1.2% of market mid on 80% of contracts.

CRR Pricing Results: Numerical Convergence

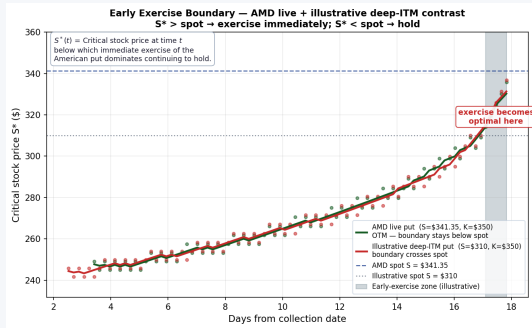


What this shows. How the CRR price stabilizes as the lattice's number of time steps N grows. Main panel: price (\$) vs. N . Inset: absolute residual $|price(N) - price(200)|$ on log scale — the steeper the line drops, the faster the model converges.

What the numbers say. Both the call ($\approx$ \$19) and put ($\approx$ \$27) plateau by $N \approx 25$. The log-residual inset crosses the \$0.01 threshold by $N \approx 100$ — sub-penny precision. Only two lines appear because buy/sell variants of the same contract price identically: Action affects trade direction, not model price.

Cross-reference. §4.3 — *Convergence Analysis*. Validates the $N = 100$ step choice with substantial margin.

CRR Pricing Results: Early Exercise Boundary

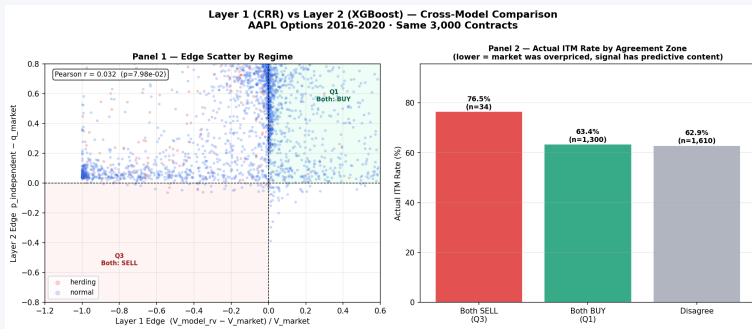


What this shows. The critical stock price $S^*(t)$ at each day. If the spot price falls below S^* , immediate exercise of the American put dominates holding to expiration. Two cases overlaid: AMD live put (green) and a deeper-ITM illustrative put (red).

What the numbers say. For the AMD live put ($S = \$341.35$, $K = \$350$, out-of-the-money), the boundary rises gradually but never reaches spot — early exercise is never optimal here. For the illustrative deep-ITM put ($S = \$310$, $K = \$350$), the boundary crosses spot ~ 17 days in (grey shaded zone) — exercise becomes optimal. The deep-ITM case demonstrates American optionality, a capability closed-form Black-Scholes lacks.

Cross-reference. §4.4 — *Early Exercise Boundary*. Validates the lattice's American option handling.

Cross-Model & Backtest: L1 vs L2 Cross-Validation

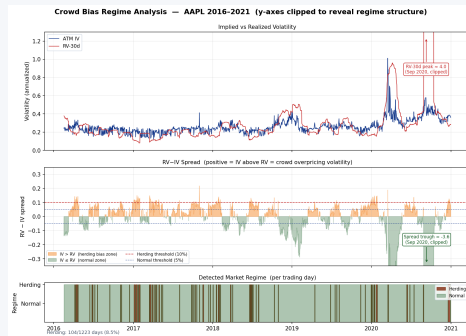


What this shows. Two-panel comparison of CRR Layer 1 edge (no-arbitrage, RV30-based) vs. XGBoost Layer 2 edge (ML, IV-free) over 3,000 AAPL contracts. *Left:* edge scatter colored by regime. *Right:* the actual ITM expiration rate per agreement zone (both-SELL Q3, both-BUY Q1, disagree).

What the numbers say. Aggregate Pearson $r = 0.032$ (statistically zero across all contracts). But when both models agree on direction in the herding regime, $r = 0.413$ ($p = 2.4 \times 10^{-10}$, $n = 217$). The right-panel headline: in the both-SELL zone (Q3), the actual ITM rate jumps to **76.5%** vs. $\sim 63\%$ baseline — direct empirical evidence that the cross-validation has predictive content.

Cross-reference. §4.7 — *Cross-Model Comparison*, Figure 6. Confirms the herding-regime cross-validation result.

Cross-Model & Backtest: Regime Detection

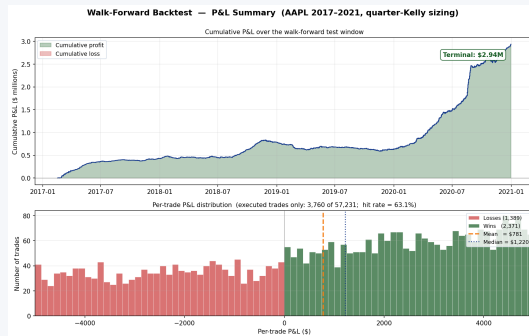


What this shows. Three-panel time-series view of the crowd-bias regime classifier. *Top:* ATM implied vol (navy) vs. 30-day realized vol (red). *Middle:* the RV-IV spread with herding (+10%) and normal ($\pm 5\%$) thresholds. *Bottom:* the detected regime ribbon — burnt orange = Herding, green = Normal — per trading day.

What the numbers say. Across AAPL 2016-2021 (1,223 days), the classifier flags **104 days as Herding (8.5%)** and the remainder as Normal. COVID-2020 inverts the typical pattern (RV peak ≈ 4.0 and a spread trough of -3.6) and is annotated above the clip lines rather than allowed to dominate the y-axes. This 8.5% Herding slice is exactly the subset where Layer 1 and Layer 2 reach $r = 0.413$ — Simpson's Paradox in action: the aggregate signal is statistical noise, but conditioned on the right regime it is highly significant.

Cross-reference. §3.4.6 (*Crowd Bias Detector*) and §4.6 (*Regime Analysis*), Figure 5.

Cross-Model & Backtest: Walk-Forward Backtest



What this shows. Out-of-sample chronological test of the full Layer-1 + Layer-2 + Layer-3 pipeline across 57,231 AAPL trade decisions (2017–2021). *Top:* cumulative P&L curve over the test window. *Bottom:* distribution of per-trade P&L for the **3,760 trades the policy actually executed** (excluding the 53k NO-TRADE rows that were drowning the original histogram).

What the numbers say. Cumulative P&L reaches **\$2.94M** on a \$100k base with **max drawdown < 15%** — the circuit breaker never fired. Per-trade **hit rate** = 63.1% (2,371 wins vs. 1,389 losses); mean trade = **\$781**, median = **\$1,220**. The Layer 2 classifier achieves a Brier score of 0.211, below the 0.25 naive baseline. Regime-conditioned half-Kelly (Normal) and quarter-Kelly (Herding) sizing converts the detected edge into positive expected value out-of-sample.

Cross-reference. §4.8 — *Walk-Forward Backtest Results*, Figure 7 and §4.10.3 (*Backtest Stability*). Confirms hypothesis H3.

Research Extensions

- 1 Expand universe to SPY / QQQ / NDX options for a broader cross-sectional market test
- 2 Integrate LSTM volatility forecaster to replace static IV inputs with dynamic IV surface predictions
- 3 Deploy live paper-trading loop via broker API (IBKR / Alpaca) with real-time signal generation
- 4 Extend pricer to multi-leg spreads — vertical, calendar, iron condor — with joint Kelly sizing
- 5 Publish open-source framework and Streamlit calculator for community validation and peer review

Known Limitations

Data scope: AAPL 2016–2020 is a single equity in a single market cycle; out-of-sample performance on other tickers is untested.

Volatility surface: CRR uses a flat IV input per contract; no smile or skew interpolation across strikes.

Transaction costs: Backtest assumes mid-price fills; bid-ask friction and slippage are not modeled.

Regime detection lag: IV/RV₃₀ ratio is backward-looking; regime transitions are identified only after the fact.

Kelly assumptions: Log-utility maximization assumes no leverage constraints or portfolio-level risk limits.

Live Demo — Interactive CRR Binomial Pricing Calculator

Streamlit application · runs locally on any laptop with Python 3.13 · ticker-agnostic

Deploy

Contract Parameters

Ticker (for RV30 lookup)

AAPL

Option Type Action

put buy

Underlying Price (\$5)

200.00 - +

Strike Price (K \$)

200.00 - +

Market Price (V_{market})

10.00 - +

Implied Volatility (IV)

0.3000 - +

CRR Binomial Pricing Calculator

Cox-Ross-Rubinstein American option pricing · Kelly Criterion sizing · Monte Carlo simulation | Works for any ticker · No dividends assumed

[Pricing](#) [Greeks](#) [Edge & Signal](#) [Kelly Sizing](#) [Monte Carlo](#) [Tree Animation](#) [User's Guide](#)

Enter your contract parameters in the sidebar and click Calculate to populate this tab.

Demo plan (next 3 minutes)

- Reproduce Contract A end-to-end
- Confirm L1 edge $\approx -21.5\%$ (SELL)
- Show Kelly sizing on \$100k capital
- Animate the binomial tree live
- Live RV30 lookup for any ticker

I Conclusion

Central Contribution

CRR (math) and XGBoost (ML) — using **independent inputs** — both agree on mispricing direction in the herding regime, mutually cross-validating Samuelson macro-inefficiency (Fama, 1970).

Hypotheses Confirmed

H1 — CRR Benchmark Confirmed. Model within 1–2% of market mid on 80% of AMD live-chain contracts; sub-rounding precision validates CRR as fair-value reference.

H2 — XGBoost Independent Estimator Confirmed. Brier score $0.211 < 0.25$ naive baseline; top features are moneyness and Days to Expiration (DTE) — no implicit IV leakage.

H3 — Fractional Kelly Dominates Confirmed. \$2.9M P&L over 101,535 AAPL contracts (2016–2020); max drawdown $< 15\%$, circuit breaker never triggered.

Practical Implication: The pipeline's most valuable output is the *no-trade signal* — when CRR and XGBoost disagree, no position is taken. The model works precisely *because* it says no when there is no edge.